



HTTP Request & Response

HTTP Request

```
[REQUEST LINE]\r\n[HEADER 1]: [value]\r\n[HEADER 2]: [value]\r\n[HEADER N]: [value]\r\n\r\n[BODY (optional)]
```

- Request line:

```
GET /index.html HTTP/1.1\r\n
```

- Method: POST, GET, PUT, DELETE
 - Target: "/index.html", "/users?id=42"
 - Version: "HTTP/1.1"
- Headers:
 - Key-value pairs in format "Key: value"

```
Host: localhost:8080  
Content-Length: 128  
Content-Type: text/plain  
Transfer-Encoding: chunked
```

- Body:

- Present for POST, PUT, and sometimes GET
- Form data: name=john&age=25
- File upload: binary or multipart data
- JSON/XML: API data

HTTP Response

```
[STATUS LINE]\r\n
[HEADER 1]: [value]\r\n
[HEADER 2]: [value]\r\n
\r\n
[BODY]
```

- Status line:

```
HTTP/1.1 200 OK\r\n
```

- Version: "HTTP/1.1"
 - Status code: 200
 - Reason Phrase (human readable description): OK
- Response headers:
 - Content-Type: text/html
 - Content-Length: 1024
 - Location: /new-page
 - For 301 redirects
 - Connection: close
 - After error or if client asks

